LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 4

Course Code: CSE252	Unit: 4
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING	Unit Title: Unsupervised Learning
Course Outcome: CO4	Unit Mapped RBT: L3 (Apply)
Paper: Sample Subjective Questions	Total Questions: 10
Maximum Marks: 10 x 10 = 100	Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply the k-Means algorithm to a table of readings collected from a robot's sensors. (10 Marks)

(a) State the preparation the readings need and the steps the algorithm performs.

(b) Use a named criterion to choose the number of clusters, and state what it contributes.

[L3 - Apply]

Q2. Compare k-Means with DBSCAN as methods of clustering. (10 Marks)

(a) State what each of the two requires before it runs and the shapes of cluster each can recover.

(b) Use a dataset containing irregular shapes and stray readings to select one of the two, with the reason.

[L3 - Apply]

Q3. Examine a clustering whose groups turn out to differ on one feature alone. (10 Marks)

(a) State the most likely cause and explain how it produces this result.

(b) Use the correction to describe what you would do, and one check you would run afterwards.

[L3 - Apply]

Q4. Use Principal Component Analysis to reduce a dataset of sensor channels before it is clustered. (10 Marks)

(a) State what the components are and the order in which they are produced.

(b) Use the explained variance to describe how many you would keep, and what you would report alongside.

[L3 - Apply]

Q5. Apply association rule mining to a supermarket's basket data. (10 Marks)

(a) State the steps from the raw baskets through to a set of rules.

(b) Use support, confidence and lift to describe how you would reduce many rules to a useful few.

[L3 - Apply]

Q6. Compare hierarchical clustering with k-Means. (10 Marks)

(a) State how each of the two forms its clusters and whether the number must be given in advance.

(b) Use the cost of each to state which suits a dataset of half a million records, with the reason.

[L3 - Apply]

Q7. Examine what support, confidence and lift each tell you about an association rule. (10 Marks)

(a) State what each of the three is computed from and what it measures.

(b) Use a rule with high confidence and a lift close to one to show why confidence alone is misleading.

[L3 - Apply]

Q8. Use the elbow method to choose the number of clusters, and state its weakness. (10 Marks)

(a) State what is plotted and how the value of k is read from the plot.

(b) Use a curve with no clear bend to explain what you would do instead.

[L3 - Apply]

Q9. Apply unsupervised learning to the sensor data of an industrial machine. (10 Marks)

(a) State two unsupervised tasks that would be useful here and what each would produce.

(b) Use one of the two to describe how the result would be checked before it is acted upon.

[L3 - Apply]

Q10. Compare dimensionality reduction with feature selection as ways of narrowing a wide dataset. (10 Marks)

(a) State what each of the two produces from the original columns, naming one method of each kind.

(b) Use a case in which every retained feature must be explained to select one of the two, and justify it.

[L3 - Apply]

----- End of Sample Question Set -----